

Kwan Moon

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U.S. CITIZEN

SUMMARY

Versatile engineer with a master's degree in computer science and extensive experience in the aerospace sector. Specialized in structural analysis and committed to leveraging analytical rigor and modern software practices to build high-performance applications.

EDUCATION

UNIVERSITY OF PENNSYLVANIA, Philadelphia, PA

Aug. 2020 – Aug. 2024

- Master of Computer and Information Technology (Computer Science)

GEORGIA INSTITUTE OF TECHNOLOGY, Atlanta, GA

Aug. 2010 – Dec. 2013

- Bachelor of Science with Highest Honors in Mechanical Engineering

INDUSTRY EXPERIENCE

Cox Automotive, Atlanta, GA

Software Engineer

Dec. 2025 – Present

- Develop features and maintain scalable and reliable backend services and APIs using Java and AWS in agile environment
- Automate GitHub Actions workflows by writing shell scripts

Lockheed Martin Corporation, Marietta, GA

Structural Analysis Engineer

Dec. 2019 – Jan. 2023

- Performed structural analysis by utilizing hand calculation methods and Finite Element Analysis tools to validate and substantiate preliminary and detailed analysis

The Boeing Company

Structural Analysis Engineer, Tukwila, WA

March 2018 - Oct. 2019

- Processed flight data and generated a log to a customer that owned Boeing military derivative platforms as part of the work scope for ASIP (Aircraft Structural Integrity Program) team to track damage accrual and fatigue life
- Worked as a cross-functional team member to track schedule, coordinate/perform tasks, discussion with a customer to support the ongoing new project
- Worked in performing fatigue analysis to substantiate a design change that affected primary structures as part of the sustaining work statement
- Worked in performing damage tolerance analysis to re-analyze various parts of Horizontal Stabilizer by using the updated fatigue spectrum to substantiate the validity of the prior existing inspection intervals and methods and update if necessary

Structural Analysis Engineer, North Charleston, SC

March 2016 - March 2018

- Performed basic structural analysis (static and fatigue for metallic and composite) for Final Assembly and Delivery of the 787 Dreamliner for various primary and secondary aircraft structures as a stress liaison to support the production in the factory by substantiating the strength check notes with new margins for affected strength checks and proposing repair options to meet the regulatory agencies requirements
- Supported urgent jobs in flight line by reviewing and providing the best feasible repair option in a timely manner to meet the structural integrity before the aircraft was delivered to the customers
- Supported systems structural projects by substantiating analyses of a design/component change and providing inputs when an initial design change needed adjustment

Structural Analysis Engineer, Seal Beach, CA

May 2014 - March 2016

- Performed basic structural analysis (static and fatigue) of wing/empennage for in-service 747 and 757 airplanes to substantiate for customers and regulatory agencies that requirements were met and satisfied
- Supported urgent repair requests (AOG: Aircraft On Ground) from airlines by reviewing and providing a feasible repair option in a timely manner to help the airplane get back to scheduled flights
- Created process modules with the intent to transfer knowledge to current and future employees in the group to advise the current practices and regulations and help understand better in order to perform tasks more efficiently

McNeil Consumer Healthcare, Fort Washington, PA

Research & Development Co-op

June 2012 - Dec. 2012

- Researched on emulsion to familiarize with theories and concepts by learning independently through online literature resources, pharmaceutical textbooks, and through the mentor
- Conducted small scale hands-on experiments as a formulator to manufacture topical pain relief cream to strengthen the robustness of the formula by generating multiple formulas to go through trial and error to inhibit the separation of emulsion within the product as much as possible to compete with the competitor's products in the future
- Performed stability studies for the multiple batches by placing a sample of each in the lab oven to observe the samples and record the progress of the stability of the sample by monitoring the duration and at what temperature the separation occurred
- Assisted marketing survey such as the preference of the shape/size for the pharmaceutical pills within the company where the inputs and data were potentially incorporated into the new products in the future

Johnson & Johnson, Lititz, PA

Process engineer Co-op

May 2011 – Dec. 2011

- Executed process validation protocols for new equipment to ensure all the requirements were met in order to manufacture new products on-site in the future
- Executed cleaning validation protocols for existing equipment to comply with regulations and maintain high quality so that there was no microbial and chemical contamination before manufacturing new batch of the pharmaceutical products in the plant
- Supervised contractors to fix the inconsistency in welding for the new product manufacturing lines by individually inspecting the welding and making a report

RESEARCH EXPERIENCE

NSF Research Experience for Undergraduates, Multi-physics of Active Systems and Structures

Department of Mechanical Engineering, Florida State University, Tallahassee, FL

Undergraduate Research Fellow

Summer 2013

- Conducted research on climbing robotic platforms
- Prototyped steering mechanisms for maneuverability on planar surfaces
- Modified an existing bipedal dynamic climbing robotic platform to climb cylindrical surfaces through use of kinematic analysis and by revisiting and implementing an active tail mechanism developed previously
- Wrote a technical research paper as a research team based on the accomplishments for the summer

SKILLS

Software:	Basic knowledge of Mathematica, Microsoft SQL Server, some experience with AUTOCAD, MATLAB and 3-D Computer Aided Design tools (CATIA V5, Dassault 3DX), ENOVIA, Finite Element Modeling (FEM) with NASTRAN/PATRAN, Altair Hypermesh, MSC APEX, Some experience with Java, Python, C, Git/GitHub, AWS, Terraform Limited experience with C++, SQLite, Docker
Testing & QA:	Unit Testing, Integration Testing, Mockito, Postman
AI Coding Assistants:	Claude Code, GitHub Copilot
Foreign Language:	Korean
Technical:	Airframe stress analysis, classical hand analysis, experience in technical writing and presentation, some experience with composites analysis, limited experience with Durability/Fatigue and Damage Tolerance, CI/CD pipelines, Microservices, REST APIs
Miscellaneous:	Experience in project management, validation and verification, manufacturing, GMP

CERTIFICATES

JavaScript Algorithms and Data Structures Masterclass, Udemy

Dec. 2021

The Complete 2021 Web Development Bootcamp, Udemy

Oct. 2021

ACTIVITIES

Faculty Honors at Georgia Tech

Fall 2010, Spring 2011, 2012, 2013

Georgia Tech Chorale, Member

Spring 2011, Fall 2013

Yesu Community Church, Treasurer

July 2010 – April 2011

- Managed financial transactions for budget of the church

Atlanta Judo Club, Member

Feb. 2007 – Nov. 2008